

Name



ID#

Math 150 Spring 2025 Final Exam

- Desmos is allowed (and the only app website allowed - no other apps or websites are allowed at anytime)
- Must show all work to receive credit
- All integrals must be
- Attach all scratch paper

	Question	Points
1	Integration	0 / 5
2	Riemann Sums	3 / 5
3	Volume	3 / 5
4	Limits & Continuity	2 / 45
5	Tangent & Normal Lines	4 / 45
6	Surface Area★	✓ 5 / 45
7	Optimization	0 / 45
8	Related Rates	- / 35
9	First Derivative Test	✓ / 35
10	Geometry	2 / 35
Total		20 / 40

Choose
one of

Graph incorrect
RtdL
was
rounded instead
of exact

1) Evaluate the integral $\int 43 \tan^5(x) \sec^4(x) dx$ using two different substitutions and discuss how the two anti-derivatives are related.

Name _____ ID# _____

2) Estimate the signed area under the curve $y = \sin\left(\frac{\pi x}{4}\right)$ from -2 to 4 using:

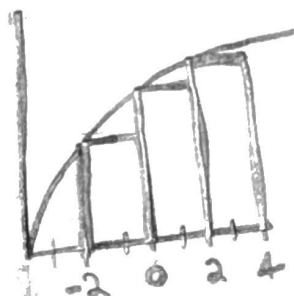
(a) three (3) rectangles with the **left endpoint** approximation.

(b) three (3) rectangles with the **right endpoint** approximation,

(c) three (3) rectangles with the **midpoint** approximation,

(d) Find the exact signed area using FTC and compare your estimates from parts (a-c)

In each case a-c draw the graph and rectangles and use sigma notation.

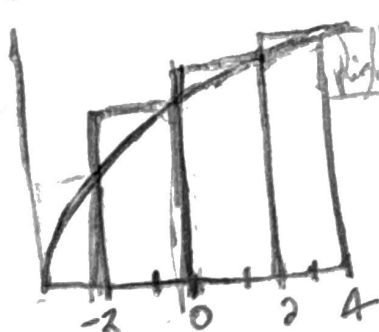


left endpoint

$$y = \sin\left(\frac{\pi x}{4}\right)$$

$$n = 3$$

$$\Delta t = \frac{4 - (-2)}{3} = \frac{6}{3} = 2$$



right endpoint

$$\sum_{k=0}^2 f(x_k) (\Delta t)$$

left point approx

$$\sum_{k=0}^2 \left(\sin\left(\frac{\pi(-2)}{4}\right) + \sin\left(\frac{\pi(0)}{4}\right) + \sin\left(\frac{\pi(2)}{4}\right) \right) (2)$$

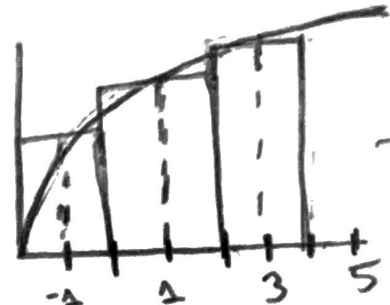
Right endpoint approx

$$\sum_{k=1}^3 \left(\sin\left(\frac{\pi(-1)}{4}\right) + \sin\left(\frac{\pi(1)}{4}\right) + \sin\left(\frac{\pi(3)}{4}\right) \right) (2)$$

Midpoint

$$\sum_{k=0}^2 \left[\sin\left(\frac{\pi(-1)}{4}\right) + \sin\left(\frac{\pi(1)}{4}\right) + \sin\left(\frac{\pi(2)}{4}\right) \right] (2)$$

FTC



Midpoint

$$\int_{-2}^4 \sin\left(\frac{\pi x}{4}\right) dx$$

$$u = \frac{\pi x}{4} \quad dx = \frac{4}{\pi} du$$

$$\int_{-\pi/2}^{\pi} \sin(u) \left(\frac{4}{\pi} du\right) = \frac{4}{\pi} \int_{-\pi/2}^{\pi} \sin u \, du$$

$$u = -\pi/2, \pi/2$$

$$\frac{4}{\pi} [-\cos u] \rightarrow \frac{4}{\pi} [-\cos \pi + \cos \frac{\pi}{2}] = \frac{4}{\pi} [-(-1) + 0]$$

$$1.273$$

$$\frac{4}{\pi}$$

3) Find the volume generated by revolving the region bounded by the curves

$$y = 2x + 4, x = 1, y = x^3$$

about the y -axis first using washers, then using shells.

Draw the region and a typical washer. Draw the region and a typical shell.



Find the exact value of volume using washers.

$$V = \pi \int_a^b [R(x)^2 - r(x)^2] dx$$

$$V = \pi \int_1^2 [(2x+4)^2 - (x^3)^2] dx$$

$$V = \pi \int_1^2 [4x^2 + 16x + 16 - x^6] dx$$

$$V = \pi \left[\frac{4x^3}{3} + 8x^2 + 16x - \frac{x^7}{7} \right] \rightarrow x=2 \text{ and } x=1$$

$$V = \frac{32}{3} + 32 + 32 - \frac{128}{7} \approx 56.381 \text{ then plug in 2}$$

Find the exact value of volume using shells.

$$V = 2\pi \int_a^b (\text{out} - \text{in}) dr$$

$$V = 2\pi \int_1^2 x [(2x+4) - x^3] dx \rightarrow 2\pi \int_1^2 x(-x^3 + 2x + 4) dx \rightarrow 2\pi \int_1^2 (-x^4 + 2x^2 + 4x) dx$$

$$\left(-\frac{x^5}{5} + \frac{2x^3}{3} + 2x^2\right) \rightarrow x=1, x=2$$

$$\left(-\frac{1}{5} + \frac{2}{3} + 2\right) = 2.467 \text{ for } x=1$$

$$\left(-\frac{32}{5} + \frac{2(8)}{3} + 2(4)\right) = 6.933$$

$$V = \frac{4}{3} + 8 + 16 - \frac{128}{7} = 25.476 \text{ then plug in 1}$$

$$56.381 - 25.476 \quad 2x+4 = x^3 \rightarrow x^3 - 2x - 4 = 0$$

$$\int x^6 = \frac{x^7}{7}$$

$$\int x^2 = \frac{x^3}{3}$$

$$\int x = \frac{x^2}{2}$$

$$\int dr = r$$

$$\frac{2^7}{7} - \frac{2(2^3)}{3} - 4$$

$$(x-2)(x^2+2x+2)$$

$$x^3 + 2x^2 + 2x - 2x^2 - 4x - 4$$

$$x^3 - 2x - 4 \checkmark$$

$$(2x+4)(2x+4)$$

$$4x^2 + 8x + 8 + 16 = 4x^2 + 16x + 24$$

$$(x^3)^2 = x^6$$

$$h = (2x+4) - x^3$$

$$6.933 - 2.467 = 4.466$$

$$2\pi(4.466) = 8.932\pi$$

Name _____ ID# _____

4) Sketch the graph of a function with following limits, and classify any discontinuities.

- $\lim_{x \rightarrow -2^-} f(x) = -2$

- $\lim_{x \rightarrow -2^+} f(x) = 1$

- $f(-2) = 3$

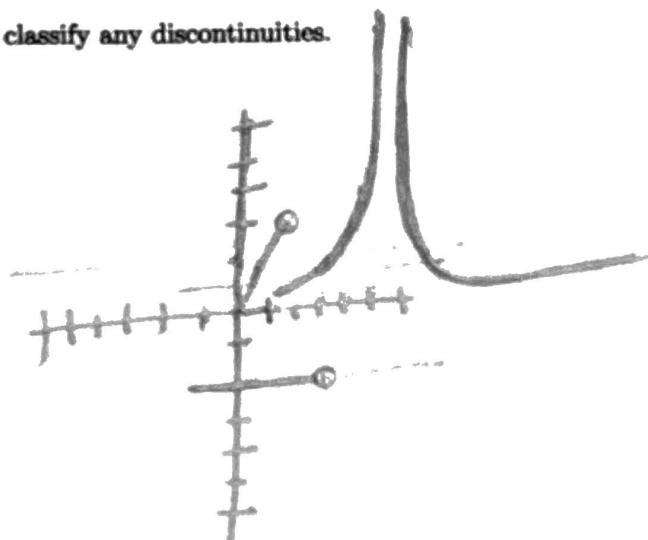
- $\lim_{x \rightarrow 3} f(x) = -2$

- $f(3) = 2$

- $\lim_{x \rightarrow 5^+} f(x) = \infty$ Infinite Discontinuity ✓

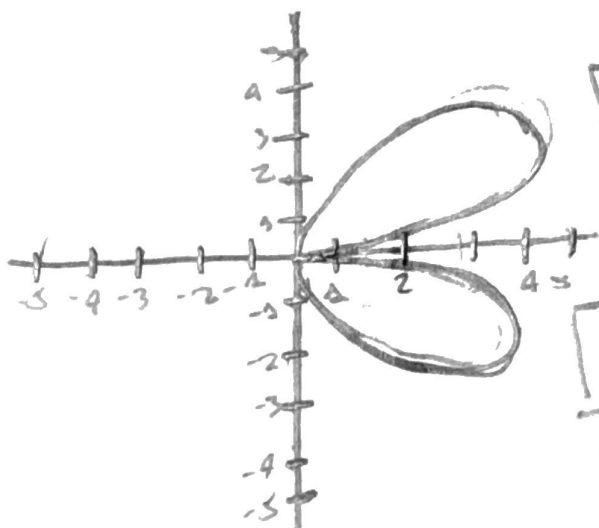
- $\lim_{x \rightarrow 5^-} f(x) = \infty$ Infinite Discontinuity ✓

- $\lim_{x \rightarrow \infty} f(x) = 1$



5) Graph the equation $12(x^2 + y^2)^2 = 169xy^2$

Find the tangent line and normal line at the point (3, 2)



Normal

$$y = -\frac{1}{1.533}(x-3)+2$$

Tangent

$$y = 1.533(x-3)+2$$

$$12(x^2 + y^2)^2 = 169xy^2$$

Chain rule

$$\frac{d}{dx} [12(x^2 + y^2)^2] = \frac{d}{dx} [169xy^2]$$

$$24(x^2 + y^2) \frac{d}{dx} (x^2 + y^2) = 169(y^2 + 2xy \frac{dy}{dx})$$

~~1.533~~

$$169 \left(\frac{d}{dx} (x) (y^2) + x \left(\frac{d}{dx} (y^2) \right) \right)$$

$$\frac{169y^2 - 48x^3 - 48xy}{48x^2y + 48y^3 - 338xy}$$

$$\frac{169(2)^2 - 48(3)^3 - 48(3)(2)}{48(3)^2(2) + 48(2)^3 - 338(3)(2)}$$

$$24(x^2 + y^2) \left(2x + 2y \frac{dy}{dx} \right) = 169(y^2 + 2xy \frac{dy}{dx})$$

$$48x^3 + 48xy + 48x^2y \frac{dy}{dx} + 48y^3 \frac{dy}{dx} = 48(x^2 + y^2) \left(x + y \frac{dy}{dx} \right)$$

$$48(x^3 + xy + x^2y \frac{dy}{dx} + y^3 \frac{dy}{dx}) = 169 + 338xy \frac{dy}{dx}$$

$$48x^2y \frac{dy}{dx} + 48y^3 \frac{dy}{dx} + 338xy \frac{dy}{dx} = -48x^3 - 48xy + 169y^2$$

$$\frac{dy}{dx} (48x^2y + 48y^3 + 338xy)$$

Name _____ ID# _____

6) Calculate the surface area of the surface obtained by revolving the curve $y = \frac{x^3}{3}$ around the x -axis from $1 \leq x \leq 2$.

$$SA = 2\pi \int_1^2 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$SA = 2\pi \int_1^2 \frac{x^3}{3} \sqrt{1 + x^4} dx$$

$$SA = \frac{2\pi}{3} \int_1^2 x^3 \sqrt{1 + x^4} dx$$

$$SA = \frac{2\pi}{3} \int_2^8 \sqrt{u} \left(\frac{1}{4} du\right)$$

$$SA = \frac{\pi}{6} \int_2^8 u^{1/2} du \rightarrow \frac{2}{3} u^{3/2}$$

~~$$\frac{2\pi}{3} \left(\frac{2}{3} \right) (17^{3/2} - 2^{3/2})$$~~

$$\frac{2}{3} (17^{3/2} - 2^{3/2})$$

$$SA = \frac{\pi}{6} \left(\frac{2}{3} \right) (17^{3/2} - 2^{3/2})$$

$$SA = \frac{\pi}{9} (17^{3/2} - 2^{3/2})$$

$$y = \frac{x^3}{3} \xrightarrow{\frac{dy}{dx}} x^2$$

$$\left(\frac{dy}{dx}\right)^2 = x^4$$

$$u = 1 + x^4$$

$$du = 4x^3 dx \rightarrow x^3 dx = \frac{1}{4} du$$

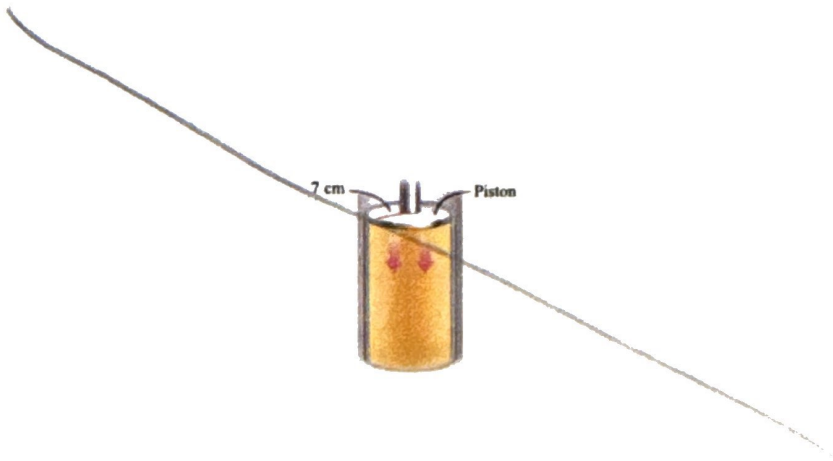
$$x=1 \rightarrow u=1+1=2$$

$$x=2 \rightarrow u=1+16=17$$

7) A wire of length 100 mm is cut into two pieces, forming a circle with the first segment, and a square with the remaining segment, while optimizing the sum of the area of the circle and square. State the objective function and the interval of interest.

Name _____ ID# _____

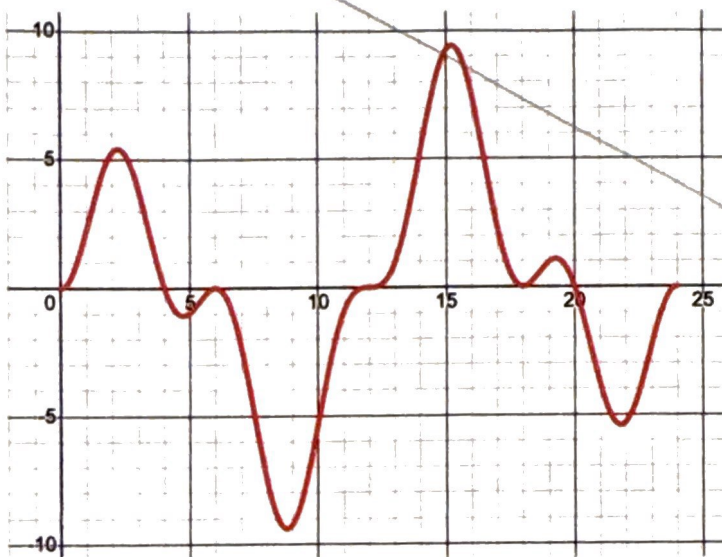
8) A piston is seated at the top of a cylindrical chamber with radius 7cm when it starts moving into the chamber at a constant speed of 3 cm / sec. (see figure).



What is the rate of change of the volume of the cylinder when the piston is 9 cm from the base of the chamber?

9) Given the graph of power $P(t)$ over a 24 hour period below, and that power $P(t) = E'(t)$.

Graph energy $E(t)$ and estimate the time(s) that **energy** is maximized and minimized.
(Hint use the first derivative test)



Name _____ ID# _____

10) Find the exact value of $\int_{-5}^0 \sqrt{25 - x^2} dx$ using geometry.

Circle $\rightarrow V = \pi r^2$

$$V = \pi(5)^2 = 25\pi$$

$$\frac{1}{4}(25\pi) \rightarrow \boxed{\frac{25\pi}{4}}$$



Find the exact value of $\int_{-5}^0 \sqrt{25 - x^2} dx$ using integration techniques.